

ATPL EASA STUDY GUIDE | 081 - PRINCIPLES OF FLIGHT

Chapter 1: Subsonic Airflow, Continuity & Bernoulli's Principle

Syllabus Source: CAE Oxford (pp. 1-34) | Pass Target: >= 90% AviationExam

1. AIR PROPERTIES, ATMOSPHERE & THE SPEED OF SOUND

Aerodynamics investigates the interaction between moving air and solid airframe bodies. Air behaves as an ideal compressible gas:

- Air Density (ρ): Mass per unit volume (kg/m^3). Standard sea level density (ISA ρ_0) = 1.225 kg/m^3 . Density decreases with increasing altitude, increasing temperature, and increasing relative humidity (humid air is LESS DENSE than dry air because H_2O molecular weight [18] is less than N_2 [28] and O_2 [32])!
- Static Pressure (p): The weight of the atmospheric column acting equally in all directions. Standard MSL pressure = $1013.25 \text{ hPa} = 101,325 \text{ N/m}^2$.
- Speed of Sound (a): The propagation velocity of pressure waves: $a = \sqrt{\gamma \times R \times T} = 38.94 \times \sqrt{T \text{ [Kelvin]}}$. The speed of sound depends EXCLUSIVELY on absolute temperature! At standard ISA sea level ($+15^\circ\text{C} / 288 \text{ K}$): $a = 661.5 \text{ kt} = 340 \text{ m/s}$.

2. THE CONTINUITY EQUATION & STREAMTUBE FLOW

For steady, continuous fluid flow through a streamtube, the principle of conservation of mass dictates that mass flow rate ($\dot{m}$) is constant:

- Continuity Equation: Mass Flow Rate: $\dot{m} = \rho \times A \times V = \text{Constant}$ (where A = cross-sectional area, V = flow velocity).
- Incompressible Flow (Subsonic Mach < 0.3): Air density is assumed constant ($\rho = \text{constant}$). Therefore: Area $\times$ Velocity = Constant: $A_1 \times V_1 = A_2 \times V_2$. As a streamtube CONVERGES (area decreases), velocity must INCREASE. As it DIVERGES (area expands), velocity must DECREASE.

3. BERNOULLI'S THEOREM & VENTURI EFFECT

Bernoulli's theorem represents the conservation of energy in steady, incompressible, frictionless streamline flow:

- Total Pressure (Stagnation Pressure): Total Pressure (P_t) = Static Pressure (P_s) + Dynamic Pressure (q) = Constant.
- Dynamic Pressure Formula: $q = \frac{1}{2} \rho V^2$. Dynamic pressure represents the kinetic energy of the moving fluid.
- Venturi Throat Dynamics: As air flows into a convergent throat: Area narrows $\rightarrow$ Velocity increases $\rightarrow$ Dynamic pressure increases $\rightarrow$ Static pressure DROPS! Downstream divergent section: Area expands $\rightarrow$ Velocity decreases $\rightarrow$ Static pressure RISES.

Regime / Duct Section	Cross-Sectional Area (A)	Velocity (V)	Dynamic Pressure (q)	Static Pressure (P_s)
Subsonic Convergent Duct	Decreases	INCREASES	INCREASES	DECREASES (Bernoulli drop)
Subsonic Divergent Duct	Increases	DECREASES	DECREASES	INCREASES (Diffuser action)
Supersonic Convergent Duct	Decreases	DECREASES	DECREASES	INCREASES (Compressibility dominates)

Supersonic Divergent Duct	Increases	INCREASES	INCREASES	DECREASES (Nozzle expansion)
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[!] EXAM TRAP (AVIATIONEXAM) - Air Density vs Humidity Exam Trap

- QUESTION: Does high relative humidity increase or decrease air density? - ANSWER: High humidity DECREASES air density! - Water vapor (H₂O) has a molecular weight of 18 g/mol, whereas dry air (N₂ and O₂) averages 29 g/mol. Humid air is lighter and less dense than dry air at the same temperature and pressure, reducing aerodynamic lift and engine thrust!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)**[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Speed of Sound & Humidity**

Step 1 (Speed of Sound): To find local acoustic speed 'a' at any flight level, calculate OAT in Kelvin ($K = C + 273$). Apply $a = 38.94 \times \sqrt{T_K}$. Example: at -56.5 C (216.5 K), $a = 38.94 \times \sqrt{216.5} = 573$ kt. Step 2 (Humidity Trap): Humid air is LESS DENSE than dry air. Water vapor (H₂O = 18 g/mol) is lighter than nitrogen (28) and oxygen (32). High temperature and high humidity reduce density, requiring longer takeoff runs.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: According to Bernoulli's equation for steady incompressible flow, what happens to static pressure when a**

- [A]: Static pressure increases
- [B]: Static pressure DECREASES while dynamic pressure increases, keeping total pressure constant
- [C]: Total pressure drops to zero
- [D]: Air density increases

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. By Bernoulli's principle, Total Pressure = Static + Dynamic. When velocity accelerates, dynamic pressure ($\frac{1}{2} \rho V^2$) rises; static pressure must decrease by an equal amount to conserve energy.

Q2: On what single atmospheric variable does the local speed of sound in air depend?

- [A]: Static pressure only
- [B]: Air density only
- [C]: Absolute Air Temperature (Kelvin) ONLY
- [D]: Relative humidity

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [C]. Formula: $a = \sqrt{\gamma \times R \times T} = 38.94 \times \sqrt{T}$. The speed of sound depends exclusively on absolute air temperature in Kelvin. Pressure and density changes cancel each other out.

Q3: How does high relative atmospheric humidity affect aircraft aerodynamic take-off performance?

- [A]: Increases lift and improves engine power

- [B]: Reduces air density, which reduces aerodynamic lift and degrades engine take-off thrust
- [C]: Increases the speed of sound
- [D]: Has zero effect on aviation

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Water molecules are lighter than nitrogen and oxygen molecules. Humid air reduces air density, decreasing dynamic pressure for a given speed, lengthening take-off ground run.